Spatiotemporal Analysis of Coastal Seagrass Dynamics in the Upper Florida Keys

Interim Report

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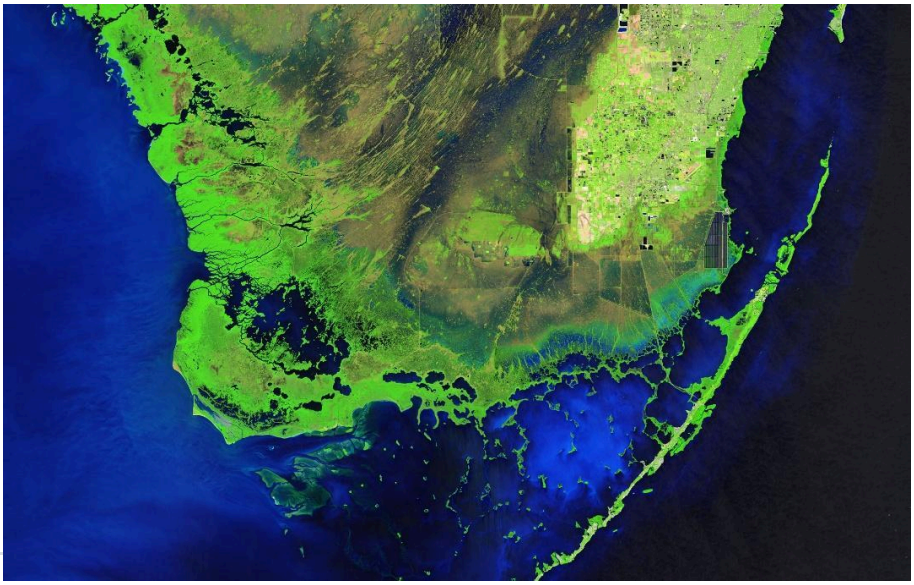
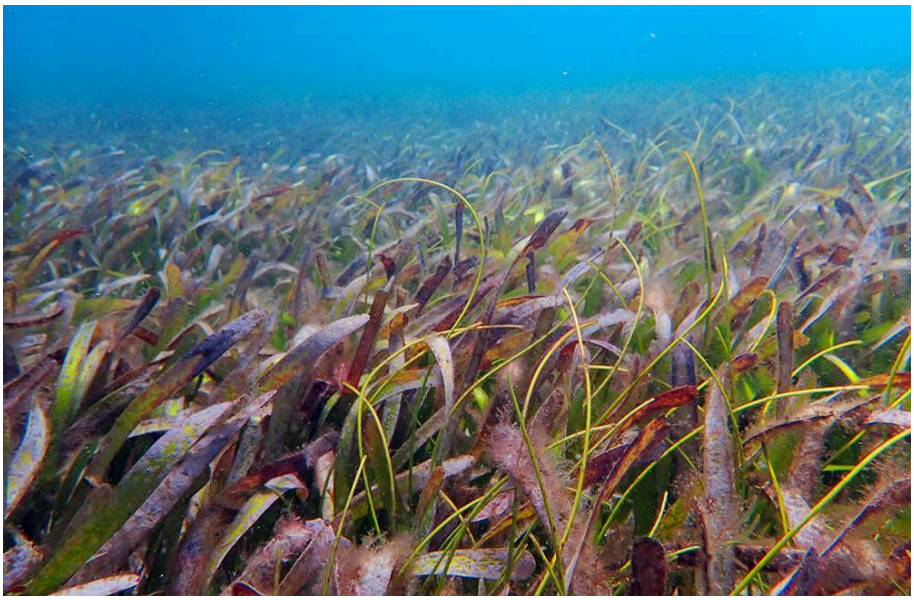
Authors: Daniel Smith, Sara Sullivan, and Omar Alazmi





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Acronyms and Abbreviations Used

S2 - Sentinel-2
FL – Florida
NOAA - National Oceanic and Atmospheric Administration
EPA - Environmental Protection Agency
USGS - United States Geological Survey
IEA - Integrated Ecosystem Assessment
GEMM - Groundwater Exchange Monitoring and Modeling plan for Central Florida Bay
FKNMS - Florida Keys National Marine Sanctuary
FIU - Florida International University
SECAR - Statewide Ecosystem Assessment of Coastal and Aquatic Resources
WQP - Water Quality Portal

Spectral Indices

NVI - Normalized Vegetation Index
NDVI - Normalized Difference Vegetation Index
NDVAI - Normalized Difference Aquatic Vegetation Index
EVI - Enhanced Vegetation Index
NDWI - Normalized Difference Water Index
NDTI - Normalized Difference Turbidity Index
DII - Depth Invariant Index

GitHub
https://github.com/ssullivan9661/seagrass-analysis/tree/main

Introduction & Background

Seagrasses are submerged flowering plants that grow in shallow coastal waters where sufficient light reaches the seafloor. Unlike seaweed (marine algae), seagrasses are flowering plants with true roots, leaves, and vascular structures that allow them to anchor in sediment and transport nutrients and water through the plant. These ecosystems form dense underwater meadows that rank among the most productive coastal ecosystems globally, providing critical nursery habitats for a wide range of marine species, including ecologically important organisms such as the spiny lobster, bonefish, and juvenile reef fishes. In addition to supporting biodiversity, seagrass meadows play a vital role in stabilizing sediments, improving water clarity through particle trapping, cycling nutrients, and sequestering carbon, making them important contributors to both coastal resilience and climate regulation.

The Upper Florida Keys represent one of many ecologically significant and environmentally sensitive coastal regions in the southeastern United States. This region forms a transitional zone between the Atlantic Ocean and the adjacent Florida Bay — Everglades estuarine system and contains extensive seagrass meadows that form the ecological foundation of shallow-water habitats. However, these systems have experienced substantial environmental stress in recent decades. A major seagrass die-off event in 2015 resulted in the loss of approximately 40,000 acres of seagrass in Florida Bay, driven by a combination of elevated turbidity, hypersalinity, nutrient enrichment, and prolonged drought conditions. This event marked a major ecological shift and provides a critical baseline for evaluating long-term ecosystem recovery and resilience.

Seagrass distribution in this region is strongly influenced by environmental conditions, particularly water clarity. Light availability is the primary limiting factor for seagrass growth, and increased turbidity caused by suspended sediments, algal blooms, and dissolved organic matter reduces light penetration to the seafloor, directly constraining photosynthesis and harming ecosystem productivity. Sentinel-2 multispectral imagery enables high-resolution observation of coastal vegetation dynamics, while in-situ water quality data allow for assessment of environmental drivers. This study integrates S2 imagery (2015–2024) with water quality observations from the Water Quality Portal (WQP) to examine how turbidity and related water quality parameters influence seagrass distribution across the Upper Florida Keys.

Objectives and Scope

This study investigates the relationship between seagrass distribution and key environmental drivers in the Upper Florida Keys over a ten-year period, with particular emphasis on water quality conditions such as salinity, temperature and turbidity, a key determinant of water clarity and light availability. The analysis integrates Sentinel-2 multispectral satellite imagery with in-situ water quality data obtained from the WQP. Spatial analysis is conducted within the Upper Florida Keys region, with temporal coverage spanning from 2015 to 2024 to capture both short-term variability and long-term environmental trends. Data processing and analysis are implemented using Python-based geospatial and numerical libraries. Results are presented through a combination of visualizations and time-series analyses to illustrate patterns

in water quality variability and seagrass distribution across the study area. Data is limited to observable environmental drivers of seagrass distribution and does not include direct field data.

Methods & Tools

This study utilizes two primary data sources: S2 multispectral satellite imagery obtained from the Copernicus Data Space Ecosystem and water quality observations collected by the USGS and EPA via the WQP. S2 offers publicly accessible satellite imagery at 10-meter spatial resolution with a five-day revisit cycle, allowing for consistent monitoring of submerged vegetation dynamics and coastal water conditions. All processing was conducted in Python using libraries such as GeoPandas and Shapely for spatial handling, and Rasterio and Rioxarray for raster processing.

Preprocessing of S2 imagery was conducted to ensure spatial and temporal consistency across the full dataset. This included scene selection, atmospheric consistency through Level-2A products, and cloud masking using the S2 Scene Classification Layer (SCL) to remove clouds, cloud shadows, and invalid pixels. Each scene was then clipped to the defined Area of Interest (AOI) using geospatial boundaries derived from a shapefile. Processing was performed on a scene-by-scene basis to maintain computational efficiency and ensure consistent handling of large datasets prior to index calculation and time-series analysis.

A set of spectral indices were calculated from S2 imagery to characterize aquatic vegetation presence and coastal water quality conditions relevant to seagrass distribution. The Normalized Difference Aquatic Vegetation Index (NDAVI) was used as the primary proxy for submerged vegetation in shallow coastal waters, providing improved sensitivity to seagrass detection compared to traditional vegetation indices. The Normalized Difference Water Index (NDWI) was applied to distinguish open water from vegetated or submerged surfaces, and the Normalized Difference Turbidity Index (NDTI) was used to estimate relative water clarity and suspended sediment concentrations, which directly influence light availability for photosynthesis. These indices were used in tandem with in-situ water quality measurements to evaluate spatial and temporal relationships between environmental conditions and seagrass distribution.

Temporal analysis was conducted to evaluate changes in seagrass-related spectral signals and associated water quality conditions over 2015–2024. Satellite-derived indices (NDAVI, NDWI, and NDTI) and in-situ water quality measurements were organized into structured time-series datasets to enable consistent comparison across annual and seasonal intervals. The Mann–Kendall trend test was applied to key environmental variables, including turbidity and selected spectral indices to identify the direction of significant temporal trends without assuming a specific data distribution. Combined with time-series visualization techniques, this allowed for the assessment of long-term changes in water quality conditions and their relationship to seagrass responses across the study period.

Implementation

The analysis will be in Python within JupyterLab notebooks hosted via Binder, utilizing libraries such as Pandas, NumPy, GeoPandas, and Rasterio to work with spatial and numerical data. With the main objectives of reproducibility and computational efficiency. To ensure organization of a clear workflow, the project follows a cookbook GitHub repository structure, to allow for easy replication of other study areas. The workflow is divided into eight notebooks numbered 00-07, starting with data acquisition (00_Copernicus_Downloading). Computational efficiency is demonstrated using a "memory-first" strategy that leverages Xarray DataArrays to handle computationally heavy S2 raster band data.

1. Data Access (Notebooks 00, 01, 04)

The workflow begins with 00 Copernicus Downloading, where S2 imagery was programmatically searched and retrieved using the simplified GeoJSON bounding box coordinates of the study area, the Upper Keys, Fl. Outline instructions for search parameters for Copernicus Data Space Ecosystem, including resolution, percent cloud cover, classification levels L1C (2015-2016) and L2A (2017-2024), due to changes in global/national product standards of S2 overtime.

Handling of S2 zip files occurs in 01 Access Data, where each scene was unzipped, and converted from JP2 → TIFF files as multi-dimensional DataArrays with Xarray, then saved as netCDF. By saving as netCDF, the project attempts to utilize lazy loading, to allow the system to easily read metadata including coordinates without pulling all 145 scenes. Simultaneously, 04 Water Quality Data queries and retrieves data from the Water Quality Portal, organized and filtered for validated historical records of turbidity, temperature, Secchi Depth, and Chlorophyll-a. The combined structuring process prepares the data in an easy format for use in later notebooks.

2. Preprocessing of S2 Imagery (Notebooks 02, 03)

This study will focus on the Upper Florida Keys as the main study area. Satellite images from S2 will be used starting from 2015, which is around the time when a large seagrass die-off event occurred [7], to capture an ecological baseline. Due to the high volume of S2 data and complexity of aquatic spectral environments, a multi-stage preprocessing pipeline was developed using Rasterio, GeoPandas, and RioXarray.

Implemented in 02 Preprocess Imagery, downloaded S2 scenes are filtered by product level: Level-1C for 2015-2016 and Level-2A for 2017-2024. Reprojecting coordinate reference system (CRS) of all scenes to match that of the study area shapefile using `to_crs` in GeoPandas. The main loop, masked cloud coverage and clipped scene by scene to the Upper Keys study area, saved, stored and wiped immediately from memory. Cloud masking was applied via the Scene Classification Layer (SCL) for all L2A level scenes, specifically for the filter classes 3, 8, 9, and

10 to ensure benthic visibility. L1C does not have a SCL, setting `masked=True` handles the NoData values in the process. Once a scene is masked and clipped, it is saved and the raw version is purged from memory.

Band	**Associated Indices**	**Importance for Project**
Blue	EVI, NDAVI	Short wavelength, low absorption; Penetrates farthest in water
Green	NDWI, NDTI, SSI-II	Highly reflectance with longer wavelength, allows to see vegetation or water clarity
Red	EVI, NDVI, NDTI	Long wavelength, high absorption; Less visibility of benthos but strong chlorophyll-a absorption
NIR (Near Infrared)	EVI, NDVI, NDWI, NDAVI	High reflectance of healthy vegetation

Calculations will be executed in in 03 Calculate Indices, spectral bands are processed to calculate spectral indices as individual Xarray DataArrays [9],[10],[11].:

$$NDVI = (NIR - Red) / (NIR + RED)$$

$$NDAVI = (NIR – Blue) / (NIR + Blue)$$

$$EVI = (2.5 * (NIR - Red) / (NIR+ (6) * Red - (7.5) * Blue + 1))$$

$$NDWI = (Green - NIR) / (Green + NIR)$$

$$NDTI = (Red – Green) / (Red + Green)$$

$$SSSII = (Green - Red) / (Green + Red)$$

$$DII = ln(Green/Blue)$$

3. Exploratory and Integrated Analysis (Notebooks 05, 06)

This phase integrates the two primary datasets, S2 indices and water quality data. 05 Water Quality Results conducts an exploratory analysis of water quality parameters across the

Upper Keys over the time period 2015-2024. Utilizing Seaborn and Matplotlib to visualize seasonal distributions and spatial means.

06 Time Series Analysis, performs a temporal join of the indices summary and water quality parameter data, by matching S2 snapshots with WQP samples using a 3 day window. Long term trends of seagrass distributions detected via the Mann-Kendall trend test applied to spectral indices and water quality parameters. Utilizing various statistical methods allows for identification of significant patterns of decline or recovery across the 2015-2024 period without assuming a normal data distribution.

4. Spatial Analysis (Notebook 07)

The final implemented phase focuses on communicating visual results of visible seagrass. Developed using Cartopy, Rasterio, and Matplotlib and highlighting interactive exploration and temporal dynamics. A user input function was developed to allow for custom comparison of any spectral index with a selected water quality parameter, to save as PNG. This tool enables the visualization of these relationships within a map projection, providing a direct way to validate results of the analysis.

For visualization of the decade long changes, the final output of the workflow is a high-resolution time-series animation of the Upper Keys, visualizing seagrass dynamic changes over the time period of the project 2015-2024, with overlay of the NDAVI index which effectively shows distribution of seagrass meadows.

Challenges Faced

One of the main challenges in this project was working with S2 satellite images and making sure the data was actually clean. Even after using cloud masking, some images still had noise from clouds or the atmosphere, which could affect the results. Because of that, we had to be careful and not just trust every image. Another challenge was understanding the indices like NDAVI, NDWI, and NDTI in a water environment. It is not as simple as land vegetation because things like water depth, turbidity, and light all affect the signal. This made it harder to clearly tell where seagrass was, so we had to think more carefully when interpreting the results.

We also had some difficulty combining the satellite data with the water quality data from the monitoring stations. A big part of this was lining up the sample collection dates with the S2 five day revisit cycle. The sampling was not always done regularly, and cloud masking also reduced the number of usable satellite images. To deal with this, we matched the data using a three day window instead of the full five days. This helped give more accurate results since water conditions can change a lot over a longer time.

A big technical challenge we faced was dealing with memory and large data. Satellite images are very large, and S2 data comes as JP2 files, so working with around 145 scenes was a

lot. To handle this, we converted the files to GeoTIFF format and used Xarray to keep everything organized. Allowing scenes to be processed one scene at a time and cleared it from memory after each step so the kernel would not crash. We saved results as metadata manifest CSV files of each netCDF scene file which avoided keeping large data stored in memory of the notebook. Another issue we ran into was when calculating indices using multiple scenes. We first tried loading everything at once, but it caused errors because the time information was not consistent across all scenes. Small differences in satellite position also caused alignment problems. To fix this, we loaded each dataset one at a time, with Xarray's `open_datasets`, because `open_mfdataset` was too fragile for our S2 data.

Github repositories prefer files under 50KB for optimal rendering and to avoid issues when uploading notebooks, specifically for the S2 processing notebooks, required that we cleared all outputs several times, despite using a memory-first strategy in implementation. Some notebooks are still unable to render completely in the repo but are readable via Binder. Working with data from 2015 to 2024 was also challenging because raster data is already large and the almost decade long time period makes for a large amount of data. Keeping everything organized and ensuring a working pipeline on any computer or PC took extra time. Finally, understanding the relationships between turbidity, water clarity, and the indices was not always straightforward. Even though we saw some clear patterns, there are multiple factors affecting the results, so we had to be careful not to oversimplify what we were seeing.

Results

1. Spatial distribution of seagrass-related spectral indices

Analysis of S2 derived indices revealed clear spatial patterns in submerged vegetation conditions across the Upper Florida Keys. The Normalized Difference Aquatic Vegetation Index (NDAVI) showed higher values in shallow, coastal regions, indicating greater likelihood of seagrass presence. These areas corresponded spatially with zones of reduced turbidity indicated by the Normalized Difference Turbidity Index (NDTI), consistent with the expectation that clearer water supports greater submerged vegetation density. The Normalized Difference Water Index (NDWI) distinguished open water from submerged vegetated regions, providing clear spatial separation between benthic habitat zones and deeper channel features. Spatial mean values were computed across 145 cloud-free S2 scenes spanning 2015–2024, covering two adjacent tiles (T17RNH and T17RNJ) that together encompass the full study area extent.

2. Relationship between water quality and spectral response

Integration of in-situ water quality data from 56 monitoring stations with satellite-derived indices yielded 387 temporally matched observation pairs, identified using a ±3 day tolerance window between sampling dates and satellite overpass dates. Pearson correlation analysis on these

matched pairs revealed statistically significant relationships between certain in-situ water quality parameters and spectral indices. Turbidity showed a significant negative correlation with NDAVI ($r = -0.288$, $p < 0.001$), confirming that higher suspended sediment amounts reduce the detectability of submerged vegetation from satellite imagery. Turbidity also showed a positive correlation with NDWI ($r = 0.325$, $p < 0.001$), indicating that increased sediment amounts elevate surface reflectance in the green wavelengths that NDWI is sensitive to. Secchi Depth showed a significant positive correlation with NDAVI ($r = 0.176$, $p = 0.006$), directly supporting that improved water clarity enables greater seagrass spectral signal. Notably, water temperature demonstrated the strongest correlations in the dataset, with SSSII ($r = -0.669$, $p < 0.001$) and NDTI ($r = 0.669$, $p < 0.001$), indicating strong seasonal coupling between water temperature and the satellite's capacity to distinguish seagrass from bare sand substrate.

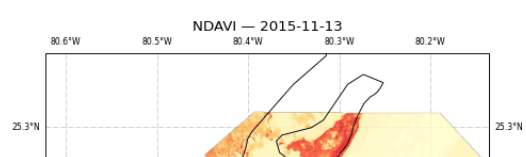
3. Temporal trends in seagrass and water quality conditions

Long-term trend analysis using the Mann–Kendall test revealed statistically significant improving trends in both water clarity and seagrass spectral signal over the study period. Secchi Depth, the primary in-situ measure of water clarity, exhibited a significantly increasing trend ($\tau = 0.163$, $p = 0.009$), indicating progressively clearer water conditions across the monitoring network. Correspondingly, NDAVI showed a significant increasing trend in satellite observations ($\tau = 0.162$, $p = 0.031$), and NDVI similarly increased significantly ($\tau = 0.153$, $p = 0.041$). However, no significant trends were detected in turbidity, salinity, or temperature over the same period, suggesting that the improving clarity signal is driven by factors beyond simple turbidity reduction alone.

4. Anomaly analysis

Quantifying monthly deviations from the 2015–2024 long-term mean revealed a transition from predominantly negative anomalies in both Secchi Depth and NDAVI during 2015–2017 to predominantly positive anomalies from 2018 onwards. The negative anomalies observed in the early study period are consistent with the degraded water quality and vegetation loss associated with the 2015 seagrass die-off event [2], suggesting that conditions in the immediate years following the die-off remained below the average in both clarity and submerged vegetation cover. Hurricane Irma's impact in September 2017 coincides with a notable inflection point in both anomaly time series, after which Secchi Depth and NDAVI anomalies shift predominantly positive and remain so through 2024. This pattern suggests the storm event may represent an ecological turning point, potentially through resuspension of nutrient-rich sediments followed by improved water column conditions during the recovery period. While causality cannot be established from the present analysis, this temporal sequence is consistent with post-disturbance recovery dynamics documented in other Florida Keys seagrass systems [7].

5. Spatial Map Analysis



The spatial maps produced highlight the changes in seagrass distributions over time with the NDAVI index overlay. The results show more visible seagrass growth over time with a sudden drop post Hurricane Irma at the end of 2017 followed by recovery in 2018. This is consistent with the findings of the computational analysis. The user function confirmed a spatial relationship between turbidity and seagrass distribution, where regions experiencing high turbidity showed less visible seagrass with the NDTI index overlay. This was accurate to the expectation.

Conclusion

Overall, our analysis confirms that seagrass distribution is strongly influenced by water clarity and environmental conditions. Spatial patterns in aquatic vegetation indices (NDAVI) show consistently higher values in regions of lower turbidity, reinforcing light availability as a primary factor for seagrass health in coastal systems. Time-series analysis suggests that seagrass health has been gradually improving since the 2015 die-off event. A positive shift in turbidity and vegetation index trends can be observed following Hurricane Irma (2017), seemingly marking an ecological turning point, possibly linked to sediment redistribution. Combining satellite-derived indices with in-situ data provides a clearer understanding of how these ecosystems change over time.

References

[1] R. J. Orth, T. J. B. Carruthers, W. C. Dennison, C. M. Duarte, J. W. Fourqurean, K. L. Heck Jr., A. R. Hughes, G. A. Kendrick, W. J. Kenworthy, S. Olyarnik, F. T. Short, M. Waycott, and S. L. Williams, "A global crisis for seagrass ecosystems," *BioScience*, vol. 56, no. 12, pp. 987–996, 2006. https://doi.org/10.1641/0006-3568(2006)56[987:AGCFSE]2.0.CO;2

[2] M. O. Hall, B. T. Furman, M. Merello, and M. J. Durako, "Recurrence of *Thalassia testudinum* seagrass die-off in Florida Bay, USA," *Marine Ecology Progress Series*, vol. 560, pp. 71–83, 2016. https://doi.org/10.3354/meps11923

[3] European Space Agency, *Sentinel-2 User Handbook*. ESA, 2015. https://sentinel.esa.int/documents/247904/685211/Sentinel-2_User_Handbook

[4] P. Villa, A. Mousivand, and M. Bresciani, “Aquatic vegetation indices assessment through radiative transfer modeling and linear mixture simulation,” *International Journal of Applied Earth Observation and Geoinformation*, vol. 30, pp. 113–127, 2014. https://doi.org/10.1016/j.jag.2014.01.017

[5] H. B. Mann, “Nonparametric tests against trend,” *Econometrica*, vol. 13, no. 3, pp. 245–259, 1945. https://doi.org/10.2307/1907187

[6] Water Quality Portal, “Water quality data,” National Water Quality Monitoring Council, 2024. https://www.waterqualitydata.us

[7] J. C. Zieman, J. W. Fourqurean, and T. A. Frankovich, “Seagrass die-off in Florida Bay: Long-term trends in abundance and growth of turtle grass, *Thalassia testudinum*,” *Estuaries*, vol. 22, no. 2B, pp. 460–470, 1999.

[8] J. W. Fourqurean and M. B. Robblee, “Florida Bay: A history of recent ecological changes,” *Estuaries*, vol. 22, no. 2B, pp. 345–357, 1999. https://doi.org/10.2307/1353063

[9] J. W. Rouse, R. H. Haas, J. A. Schell, and D. W. Deering, *Monitoring vegetation systems in the Great Plains with ERTS*. NASA Special Publication, 1974.

[10] B. C. Gao, “NDWI—A normalized difference water index for remote sensing of vegetation liquid water,” *Remote Sensing of Environment*, vol. 58, no. 3, pp. 257–266, 1996.

[11] A. I. Dogliotti, K. Ruddick, B. Nechad, et al., “A single algorithm to retrieve turbidity from remotely sensed data in all coastal and estuarine waters,” *Remote Sensing of Environment*, vol. 156, pp. 157–168, 2015.

[12] Lai, Y., Zhou, Z., Su, B., Xuanyuan, Z., Tang, J., Yan, J., Liang, W., & Chen, J. (2022, November 11). *Single underwater image enhancement based on differential attenuation compensation.* Frontiers. https://www.frontiersin.org/journals/marine-science/articles/10.3389/fmars.2022.1047053/full

[13] Torrez-Perez, J. L., McCullum, A., Beaudry, B., & Cruz, S. (2023, November 2). *Spectral indices for land and aquatic applications part 2.* NASA. https://appliedsciences.nasa.gov/sites/default/files/2023-10/Spectral_Indices_Part2.pdf

[14] David R. Lyzenga. Appl. Opt. 17, 379-383 (1978). *"Passive remote sensing techniques for mapping water depth and bottom features,". https://opg.optica.org/ao/abstract.cfm?URI=ao-17-3-379*

[15] *Rioxarray*. 13. Rioxarray - Introduction to GIS Programming. (n.d.). https://geog-312.gishub.org/book/geospatial/rioxarray.html

Data interpretation assisted by Claude

Anthropic, *Claude*, Large language model, 2026. Available: https://claude.ai/

Google. (Year). *Gemini* (Month Day version) [Large language model]. https://gemini.google.com/